

ACCEPTED MANUSCRIPT • OPEN ACCESS

All-solution-processable hydrogen-bonded organic framework artificial synapse for neuromorphic application

To cite this article before publication: Jie Tang *et al* 2026 *Int. J. Extrem. Manuf.* in press <https://doi.org/10.1088/2631-7990/ae8f85>

Manuscript version: Accepted Manuscript

Accepted Manuscript is “the version of the article accepted for publication including all changes made as a result of the peer review process, and which may also include the addition to the article by IOP Publishing of a header, an article ID, a cover sheet and/or an ‘Accepted Manuscript’ watermark, but excluding any other editing, typesetting or other changes made by IOP Publishing and/or its licensors”

This Accepted Manuscript is © 2026 The Author(s). Published by IOP Publishing Ltd on behalf of the Institute of Machinery Manufacturing Technology.



As the Version of Record of this article is going to be / has been published on a gold open access basis under a CC BY 4.0 licence, this Accepted Manuscript is available for reuse under a CC BY 4.0 licence immediately.

Everyone is permitted to use all or part of the original content in this article, provided that they adhere to all the terms of the licence <https://creativecommons.org/licenses/by/4.0>

Although reasonable endeavours have been taken to obtain all necessary permissions from third parties to include their copyrighted content within this article, their full citation and copyright line may not be present in this Accepted Manuscript version. Before using any content from this article, please refer to the Version of Record on IOPscience once published for full citation and copyright details, as permissions may be required. All third party content is fully copyright protected and is not published on a gold open access basis under a CC BY licence, unless that is specifically stated in the figure caption in the Version of Record.

View the [article online](#) for updates and enhancements.

All-solution-processable hydrogen-bonded organic framework artificial synapse for neuromorphic application

Journal:	<i>International Journal of Extreme Manufacturing</i>
Manuscript ID	IJEM-113612.R1
Manuscript Type:	Paper
Keywords:	hydrogen-bonded organic frameworks, all-solution-processable devices, artificial synapses, transient electronics, neuromorphic systems



All-solution-processable hydrogen-bonded organic framework artificial synapse for neuromorphic applications

Jie Tang^{1,2,§}, Yanqiu Sun^{2,§}, Jianlong Shao¹, Shijie Chen¹, Cheng Zhang^{1,*}, Yiming Liu³, Qifeng Lu³, Tong Tong^{1,*}, Chen Cao¹, Fangchao Li¹, Xinli Cheng¹, Shiqing Zhao¹, Qianfeng Gu⁴, Chunlan Ma¹, Qichun Zhang^{4,*}, and Yang Li^{1,*}

¹ Key Laboratory of Intelligent Optoelectronic Devices and Chips of Jiangsu Higher Education Institutions, School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou 215009, China

² School of Chemistry and Life Sciences, Suzhou University of Science and Technology, Suzhou 215009, China

³ School of CHIPS, XJTLU Entrepreneur College (Taicang), Xi'an Jiaotong-Liverpool University, Taicang 215400, China

⁴ Department of Materials Science and Engineering, Department of Chemistry & Center of Super-Diamond and Advanced Films (COSDAF), City University of Hong Kong, Kowloon, Hong Kong SAR 999077, China

§ These authors contributed equally to this work.

E-mail: zhangcheng@usts.edu.cn; tongtong@usts.edu.cn; qiczhang@cityu.edu.hk; liyang@usts.edu.cn

Abstract

Organic materials that exhibit gradient conductance plasticity are currently regarded as promising candidates to play the key role of bionic synapses in flexible neuromorphic circuits, mimicking brain-like processing behavior. However, owing to their poor long-term operating stability, low intrinsic conductivity, and inferior charge transfer ability, their electronic synaptic applications are largely restricted. Herein, we controllably synthesized a two-dimensional (2D) flake-like hydrogen-bonded organic framework composed of meso-tetra(carboxyphenyl)porphyrin monomers (TCPP-HOF), which serves as a reliable synaptic medium with high material stability and enhanced charge-carrier transportation. An all-solution-processed 10×10 2D MXene/TCPP-HOF/MXene/polyimide (PI) heterostructured flexible device array using MXene as the electrode and PI as the substrate is fabricated, presenting gradient conductance modulation under both continuous electrical scanning and pulse algorithms, closely simulating biological synaptic behaviors. More importantly, such a metal-electrode-free device is easily degradable, giving the possibility of recyclable transient electronics and information security. This work sets a precedent for the development of highly viable HOF-manipulated artificial synaptic mimicry for low-cost, easily processed, and highly efficient neuromorphic applications.

Keywords: hydrogen-bonded organic frameworks, all-solution-processable devices, artificial synapses, transient electronics, neuromorphic systems

1. Introduction

Flexible neuromorphic electronic devices for intelligent wearable systems have attracted great attention because of their advantages of energy efficiency, high speed, and in-memory computing capacity [1-3]. In the pursuit of hardware neural networks, artificial devices that fully mimic biological synapses are vital. Currently, prototypes of electronic devices with different structures, such as ferroelectric transistors, complementary metal-oxide semiconductor (CMOS) transistors, and phase change devices, have shown promise for simulating the biological neural network of synapses [4-7]. Among them, two-terminal memristive devices outperform due to their simple structures, inherent adjustable multiple resistances, and memory capabilities, which are similar to the variable synaptic weights between presynaptic and postsynaptic neurons [8]. So far, remarkable achievements have been made in bionic memristive synapses, mimicking the basic functions, including short-term plasticity (STP), long-term plasticity (LTP), and spike-timing-dependent plasticity (STDP) [9].

Currently, the synaptic devices face numerous challenges, such as high cost, complex fabrication processes, and intensive energy consumption, which arise from two main aspects: (1) the active layers, such as metal oxides and two-dimensional (2D) materials, usually require physical and/or chemical vapor deposition, making large-scale production extremely difficult and expensive; (2) for the active layers that can be prepared via solution processing (e.g., organic and organic-inorganic hybrid materials), the device electrodes still rely on vacuum deposition techniques, resulting in high cost and energy dissipation [9-13]. In addition, most existing synaptic devices are non-degradable, resulting in electronic pollution and resource waste. These challenges seriously hinder the development of synaptic devices and pose major obstacles to green electronics [14-18]. Therefore, the construction of all-solution-processed and degradable transient synaptic devices is vital. To this end, we propose an innovative comprehensive solution of employing solution-processable organic semiconductors as the active layer, highly conductive MXenes as the electrodes, and easily degradable self-supported polyimide (PI) films as the substrates, which eliminates the reliance on traditional high-energy-consuming fabrication methods and provides a strong guarantee for the environmental friendliness of intelligent devices.

In this contribution, we prepared a tetrakis(4-carboxyphenyl)porphyrin-based hydrogen-bonded organic framework (TCPP-HOF) as the organic active layer and fabricated a 10×10 2D MXene/TCPP-HOF/MXene/PI synaptic emulator. The MXene- $\text{Ti}_3\text{C}_2\text{T}_x$ array is inkjet-printed as top and bottom electrodes, whereas the TCPP-HOF is spin-coated between the electrodes as a switching element. All the device fabrication procedures are solution processable, which guarantees low cost, scalability, and ease of operation. The device exhibits gradient conductance modulation with an ultrafast switching speed of 26 ns under a continuous voltage scanning and pulse algorithm, thus approximating the essential synaptic functions and implementing

highly efficient HOF-based neuromorphic application. In addition, this device can rapidly degrade in the monomer solution of the PI film, especially under heated conditions, ensuring information security in special scenarios. Moreover, such a bionic device demonstrates convolutional neural network (CNN)-based computing with an exalted digit recognition accuracy of up to 97.23%, providing a promising paradigm for the development of green, easy-to-process, and highly efficient neuromorphic computing systems.

2. Results and Discussion

The hydrogen-bonded porphyrin framework (TCPP-HOF) was synthesized by self-assembly of 5,10,15,20-tetra(4-carboxyphenyl)porphyrin (TCPP) in a mixed solvent of N,N-dimethylformamide (DMF) and 1,2,4-trichlorobenzene (TCB) at 140 °C, as schematically illustrated in Figure 1(a) [19-21]. The theoretical molecular simulation revealed the formation of a square-layered topological TCPP-HOF structure (Figure 1(b)). Moreover, the PI film was synthesized via a condensation reaction. Following the dissolution of p-phthalaldehyde (0.02 mol) in DMF (50 mL), diethylenetriamine (0.6 mmol) was added with stirring. After thorough mixing, tris(2-aminoethyl)amine (9.4 mmol) was added dropwise. The solution was promptly transferred into a mold made of silicone-coated release paper. The film was cured through a step-wise room temperature to 80 °C thermal treatment, followed by a heat-pressing operation (Figure 1(c)) [22]. After complete evaporation of the volatiles, a transparent yellow PI film was obtained, serving as a flexible and degradable substrate.

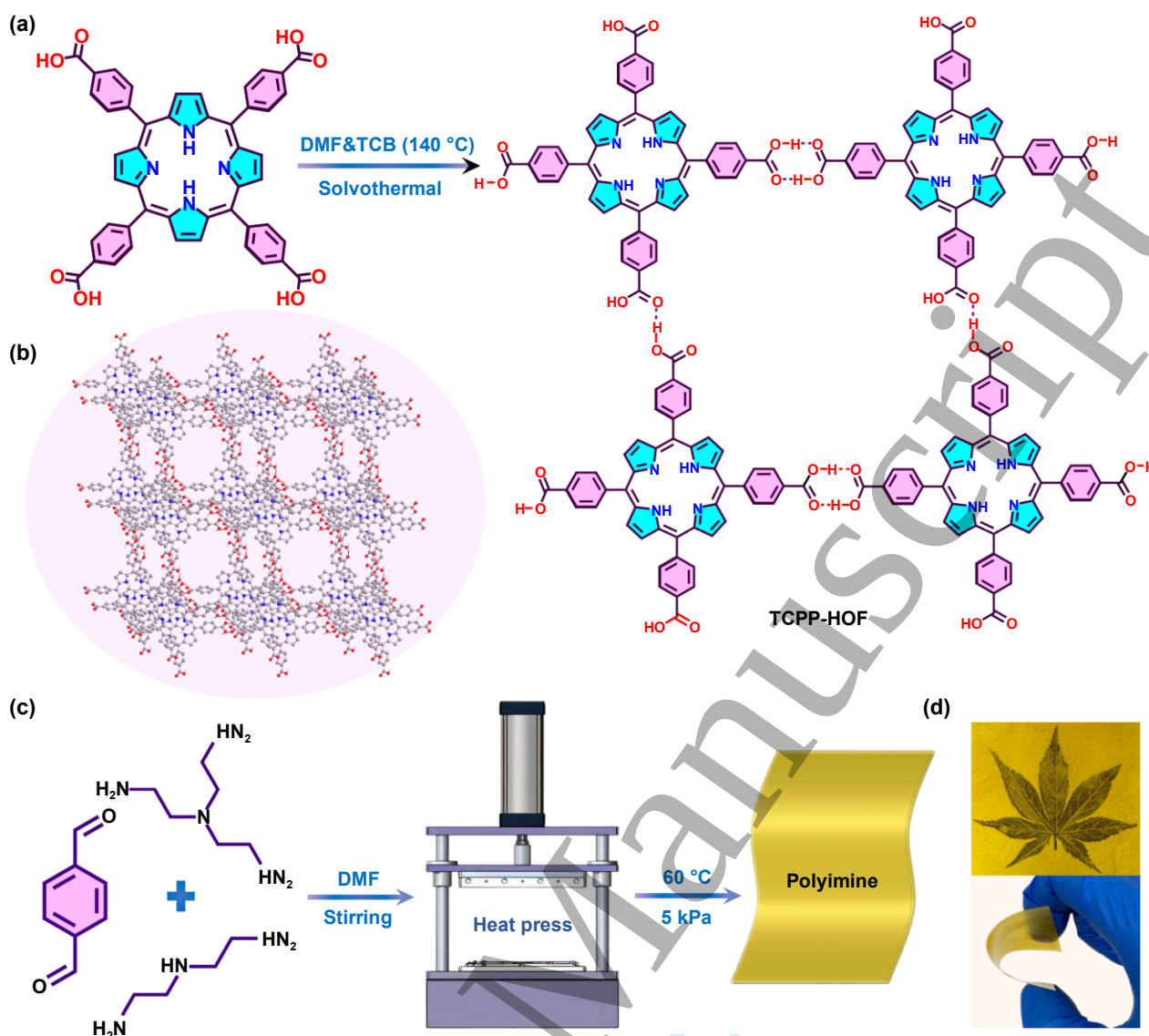


Figure 1. Synthesis of TCPP-HOF and PI film. (a) Synthetic procedure of TCPP-HOF. (b) Theoretically calculated topology of the TCPP-HOF along the channel direction. (c) Synthetic procedure of the PI film. (d) Transparency and flexibility of the PI film.

In light of the all-solution-processable design, we fabricated a 10×10 TCPP-HOF-based synaptic device array using highly conductive MXenes as the top/bottom electrodes and a degradable PI film as the substrate, affording a 2D MXene/TCPP-HOF/MXene/PI stacked structure (Figure 2(a)). Firstly, a PI film substrate ($2.0 \text{ cm} \times 2.0 \text{ cm}$) was pre-heated on a hotplate at 50°C . A solution of MXene- $\text{Ti}_3\text{C}_2\text{T}_x$ (Figure 2(b)) with a concentration of $1.0 \text{ mg} \cdot \text{mL}^{-1}$ was spray-coated onto the PI film through a strip-shaped mask. Then a TCPP-HOF solution ($1.0 \text{ mg} \cdot \text{mL}^{-1}$) was spin-coated onto the MXene electrode surface and dried at 50°C to form a uniform active layer. Finally, a second layer of MXene was evenly sprayed and dried, yielding a flexible device with a 10×10 2D MXene/TCPP-HOF/MXene/PI architecture (Figure 2(c)). Compared with traditional inorganic devices (e.g., metal oxide synapses) requiring high-vacuum and energy-intensive deposition, this all-solution-processed strategy not only simplifies the fabrication procedure but also enables low-cost manufacturing, providing a scalable route for green and printable electronics.

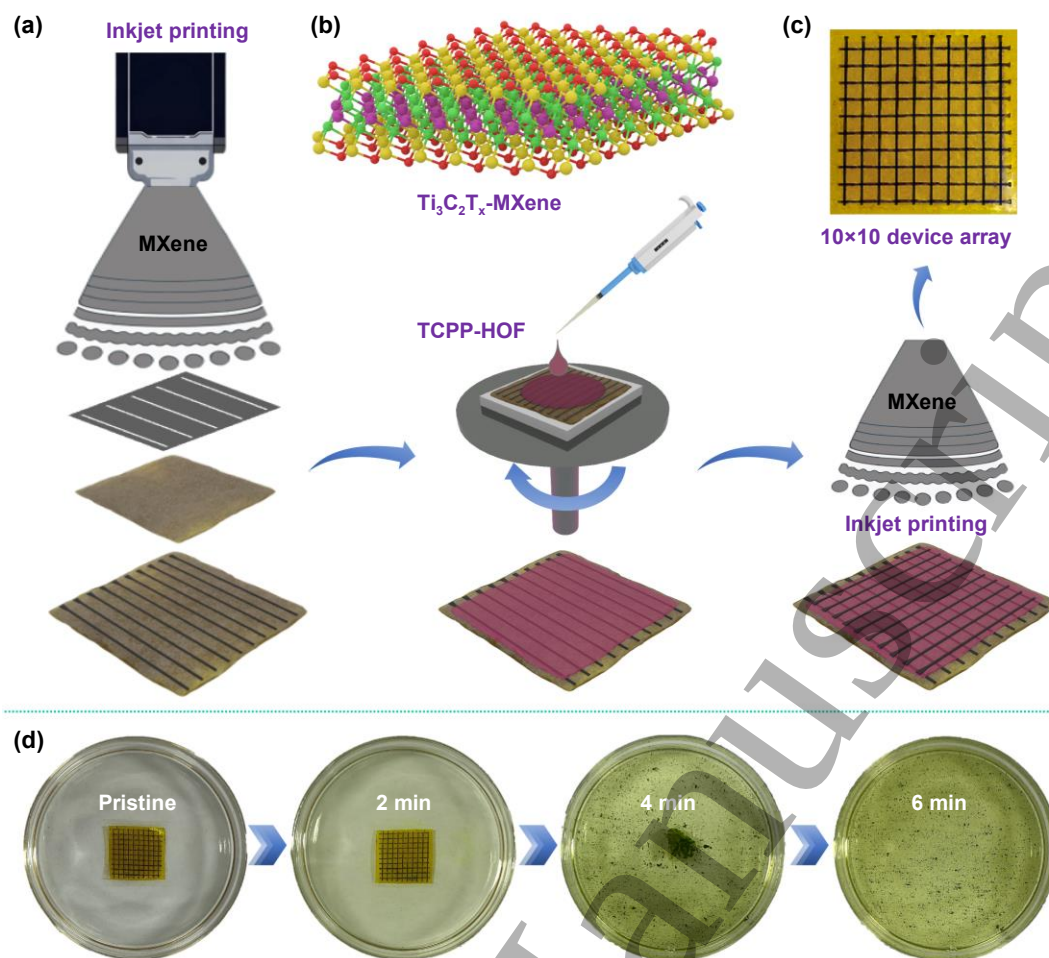


Figure 2. The device fabrication and decomposition processes. (a) The preparation flow chart of the 10×10 2D MXene/TCPP-HOF/MXene/PI synaptic device array. (b) Schematic structure of $\text{Ti}_3\text{C}_2\text{T}_x\text{-MXene}$. (c) Photograph of the 10×10 MXene/TCPP-HOF/MXene/PI device array. (d) Decomposition process of the MXene/TCPP-HOF/MXene/PI device array.

The dynamic covalent chemistry of the PI substrate enables a straightforward degradation solution that tackles electronic waste while creating opportunities for transient electronics. The degradation process is depicted in Figure 2(d). Immersing the device in a solution containing diethylenetriamine and tris(2-aminoethyl)amine under heating leads to complete decomposition into the soluble monomeric constituents of PI within 6 minutes, with any inert residues settling out [22]. This approach underscores the potential of dynamic covalent polymeric films in creating sustainable flexible electronics.

The synaptic plasticity of the MXene/TCPP-HOF/MXene/PI device was systematically evaluated to investigate the characteristic responses under a variety of stimuli, including the key functions of long-term potentiation (LTP), long-term depression (LTD), paired-pulse facilitation (PPF), and post-tetanic potentiation (PTP) [23-25]. The electrical characteristics of the synaptic device were measured through typical current–voltage (I – V) sweeps, which were recorded by voltage scanning between -1.0 V and $+1.0$ V. Figures 3(a) and S1 illustrate the cyclic I – V characteristic curves when a moderate voltage sweep path ($0 \rightarrow 1.0 \rightarrow 0 \rightarrow -1.0 \rightarrow 0$ V) was applied. From the 1st cycle to the 38th cycle, the TCPP-HOF device demonstrates gradient

conductance under continuous positive and negative voltage scans, indicating an accumulated resistance change [26]. Notably, the device maintains its I - V characteristics under heating at 120 °C and affords batch-to-batch uniformity (Figure S1(b)–(d)), indicating satisfactory thermal stability.

When subjected to successive voltage stimuli, the artificial synapse exhibits a gradual time-dependent conductance change (ΔW_i ; Figure 3(b) and (c)), which is characterized by a monotonic current increase of more than 600 s during continuous voltage sweeps at both polarities (Figure 3(d)). The device achieves stable and graded conductance modulation, which is essential for emulating the progressive variation in synaptic strength [27–28]. Notably, the temporal evolution of conductance, controlled by the cumulative voltage stimulus, can effectively mimic dynamic synaptic weight updates [26, 29]. Furthermore, the device demonstrates LTP under consecutive pulse trains. Figure 3(e) and (f) display the corresponding conductance evolution under continuous pulses with amplitudes of 1.0 V and –1.0 V, respectively (width: 203 ms; interval: 309 ms). This behavior mimics a synaptic reinforcement mechanism, wherein the repeated potentiation under a constant learning duration strengthens memory retention [30–32]. To further detect LTP, the synaptic device was subjected to extended pulse training. The cumulative influence of programming voltage pulses was examined by varying the number of applied pulses from 0 to 180 [26, 33–34]. Figure 3(g) shows the resulting ten conductance states, read at a 0.1 V bias after programming with 1.0 V pulses. These states are distinctly represented as “0–9” levels, demonstrating the ability to achieve nonvolatile conductance states [35]. Such long-term memory (LTM) characteristics highlight the device’s substantial promise for high-density information storage applications.

To emulate synaptic plasticity, we characterized the conductance dynamics under transient pulse stimuli of varying amplitudes, frequencies, and durations to identify the optimal synaptic-mimicking parameters [33]. The positive pulse trains induce an increase in the transient current across different amplitudes, widths, and intervals (Figure S2), a similar trend observed under negative pulse trains (Figure S3). Consequently, the device emulates a biological synaptic response when stimulated by programmed voltage spikes. Through systematic modulation, the optimal postsynaptic current (PSC) responses are achieved with a positive pulse (amplitude: 8.0 V; width: 1 μ s; interval: 2 μ s; Figure 3(h)) and a negative pulse (amplitude: –11.0 V; width: 1 μ s; interval: 9 μ s; Figure 3(i)). Central to the assessment of the device’s short-term plasticity (STP) are the phenomena of the PPF and PTP. Their indices, calculated via Equations (S1) and (S2) detailed in Note S1, are functions of PSCs (I_1 , I_2 , and I_{10}) triggered by the initial, second, and tenth pulses [26, 33, 36]. PPF is observed as a progressive amplification of the PSC from the first to the second paired pulse. The efficacy of this facilitation, encapsulated by the ΔW_{PPF} value, exhibits an inverse relationship with the inter-pulse interval (Δt), as shown in Figure 3(j) [36,37]. An analogous decay is noted for the ΔW_{PTP} metric with increasing Δt . Increasing these indices requires decreasing Δt or increasing the pulse number, suggesting that the temporal proximity of stimuli intensifies learning efficacy. The asymptotic behavior of ΔW_{PPF} and ΔW_{PTP} at large Δt values denotes a transition from short-term modulation to a stable state [38].

Furthermore, this device demonstrates the ability to emulate another brain-inspired function of prepulse inhibition (PPI) [26, 39]. In biological synapses, an initial stimulus can suppress responses to subsequent inputs, enabling the central nervous system to prioritize temporally salient signals. To assess this, four distinct pulse sequences (P1–P4) were applied to the device (Figure S4). The results show that a weak preceding pulse effectively inhibits the response to stronger subsequent pulses. The inhibition efficiency (η) was quantified as $\eta = (I_b - I_a)/I_a \times 100\%$, where I_a and I_b denote the current responses to the first pulse in P1 and the final pulse in each sequence, respectively. As illustrated in Figure 3(k), this PPI behavior mimics a neurobiological mechanism for alerting to potential hazards within a given time window, analogous to certain memory-based warning functions in the human brain.

Moreover, as summarized in Figure 3(l) and Table S1, our device outperforms state-of-the-art HOF-based synaptic devices across key metrics: a faster response speed of 26 ns (Figure S5), a greater number of ten conductance states, and a superior convolutional neural network (CNN)-based image recognition accuracy of 97.23% [26, 40–43]. The ultrafast switching speed of 26 ns can be attributed to efficient charge-carrier hopping within the highly oriented 2D HOF framework. This rapid response, combined with multiple stable conductance states, allows the device to process high-frequency signals with minimal latency, closely mimicking the dynamic weight tuning of biological synapses. These results collectively demonstrate the exceptional switching characteristics of TCPP-HOF-based device and significant potential for neuromorphic computing applications.

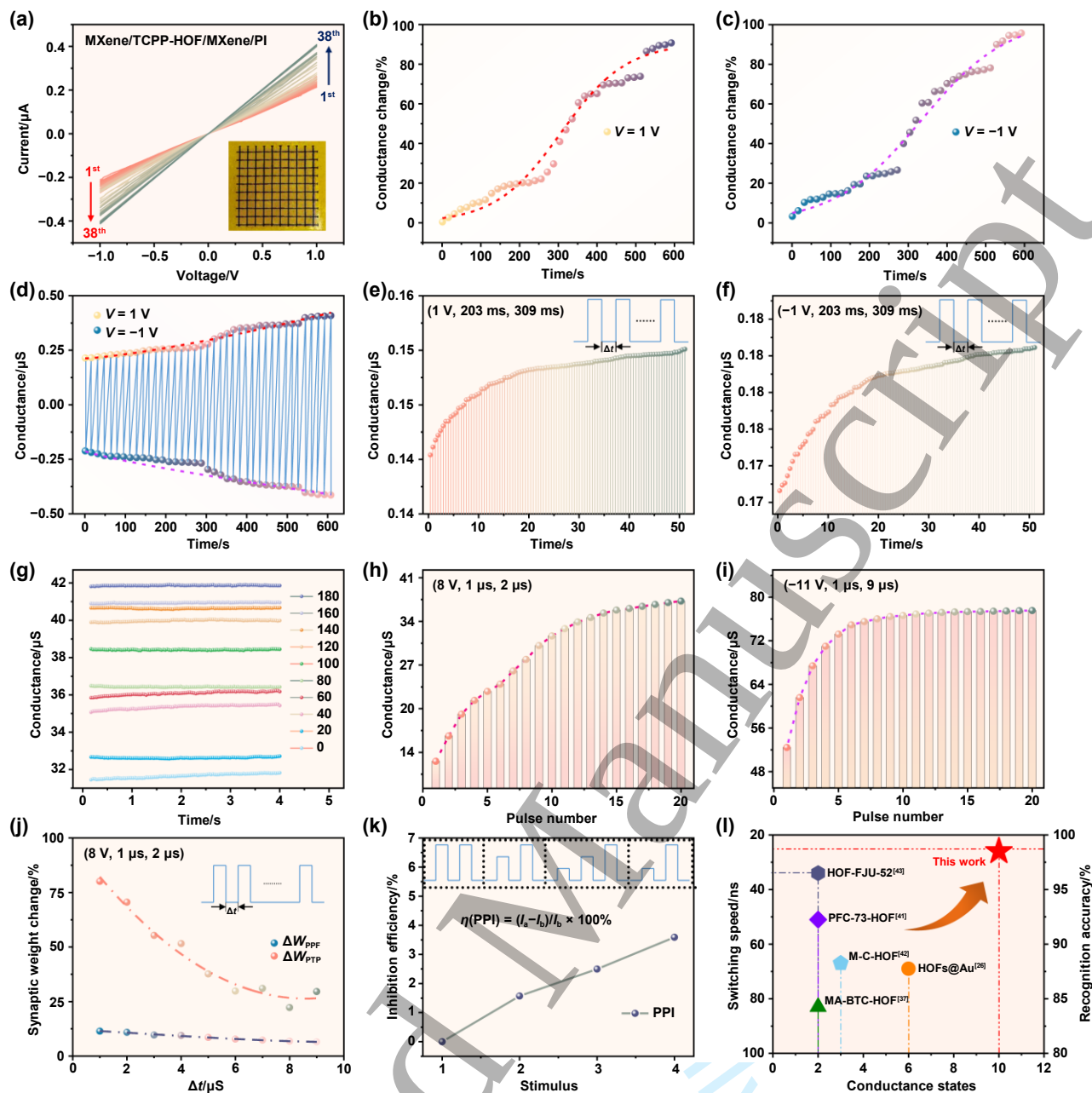


Figure 3. Synaptic behaviors of the MXene/TCPP-HOF/MXene/PI device. (a) Cyclic $I-V$ characteristics under consecutive voltage sweeps ($0 \rightarrow 1.0 \rightarrow 0 \rightarrow -1.0 \rightarrow 0$ V). (b) and (c) ΔW_i under continuous positive (b) and negative (c) voltage stimulus. (d) Current variation over time at ± 1.0 V. (e) and (f) Conductance change under continuous positive (e) and negative (f) voltage pulse stimulations. (g) Ten programmable conductance states using write pulses with a read voltage of 0.1 V. (h) and (i) Conductance response to positive (h) and negative (i) pulse trains. (j) PPF and PTP indices versus Δt . (k) PPI function. (l) Performance comparisons of the switching time, conductance states, and recognition accuracy with other reported HOF-based devices [26, 40-43].

To gain insight into the mechanism behind the synaptic behaviors, we performed detailed structural characterization and theoretical calculations. The prepared TCPP-HOF material was characterized using scanning electron microscopy (SEM), transmission electron microscopy (TEM), and X-ray diffraction (XRD).

SEM imaging reveals the successful formation of sheet-like TCPP-HOF structures through the self-assembly process (Figure 4(a)). TEM measurement indicates that the TCPP-HOF has a nanoplate-like structure (Figure 4(b)). Moreover, the high-resolution TEM (HRTEM) image manifests distinct lattice fringes with a lattice spacing of 0.327 nm. These results suggest that the ordered nanoplate-like structure of TCPP-HOF can be attributed to intermolecular interactions within its conjugated framework [44-46]. The characteristic small-angle XRD peaks of TCPP-HOF confirm the successful formation of the target structure (Figures 4(c) and S6) [19]. Notably, the π - π stacking distance determined by XRD is 0.349 nm, which is in accordance with the observed lattice spacing determined by HRTEM. Furthermore, Fourier transform infrared (FT-IR) and Raman spectroscopy collectively confirmed the successful synthesis of TCPP-HOF. FTIR analysis shows that TCPP-HOF has porphyrin fingerprint peaks at 1 730 cm^{-1} , 1 570 cm^{-1} , 1 450 cm^{-1} , 1 330 cm^{-1} , 1 190 cm^{-1} and 720 cm^{-1} , which are attributed to the stretching vibrations of carboxyl C=O, phenyl C=C, pyrrole C-N, carboxyl C-OH, imino N-H and pyrrole ring deformation, respectively (Figure 4(d)). The Raman spectrum in Figure 4(e) indicates that TCPP-HOF has characteristic vibration bands of porphyrin at 1 000 cm^{-1} (pyrrole C-H), 1 240 cm^{-1} (carboxyl C-OH), 1 320 cm^{-1} (pyrrole C-N), 1 470 cm^{-1} (C-C), and 1 560 cm^{-1} (phenyl C=C).

Since small molecular porphyrins tend to aggregate through intermolecular π - π interactions, we incorporated these functional moieties into the porous framework. Individual porphyrin units are connected to four adjacent units through O-H $\cdots$ O hydrogen bonds via carboxyl groups. Adjacent layers are stacked in the AA mode, with an interlayer spacing of 3.49 Å (Figure 4(f)). The porphyrin moiety in TCPP-HOF is determined to be non-planar and slightly bent [21]. In biological neural systems, neurotransmitters bridge neurons by crossing the synaptic cleft to transmit electrical signals from the presynaptic terminal to the postsynaptic terminal [46-47]. This process can be emulated in our synaptic device. Like neurotransmitters modulate postsynaptic potential by altering ion channel activity to enable signal propagation, the migration and accumulation of holes within the device alter its conductance, thereby achieving multilevel resistive switching [33, 46, 48]. To elucidate the underlying operating mechanism, we performed density functional theory (DFT) calculations to analyze the electrostatic potential (ESP) and frontier molecular orbitals of TCPP-HOF (Figures 4(g) and S7). The molecular ESP isosurface reveals an open channel (positive ESP, white and blue) that facilitates charge transport, alongside electron-deficient acceptor sites (negative ESP, red) that act as trapping centers to impede charge migration. The computational results further indicate that the highest occupied molecular orbital (HOMO) is primarily localized on the donor moiety, whereas the lowest unoccupied molecular orbital (LUMO) is distributed mainly on the acceptor. This spatial separation of electron density suggests the feasibility of a charge transfer (CT) process between the donor and acceptor. Under an applied voltage, charge carriers gain sufficient activation energy to be injected from the donor to the acceptor, forming a conductive pathway and triggering resistive switching. Specifically, as the positive voltage increases, the electric field-induced CT drives electrons from the donor to the acceptor, where they are trapped. This leads to electron retention and the establishment of a conductive channel, thereby increasing the conductivity. Conversely, upon the application of a negative voltage, a charge-detrapping process occurs, restoring the

device to its initial high-resistance state. Notably, the hydrogen bond-regulated HOF periodic framework provides continuous and ordered internal CT pathways. The progressive CT processes among the HOF framework enable dynamic conductance modulation and gradient plasticity, which are crucial for mimicking synaptic weight updates in neuromorphic systems.

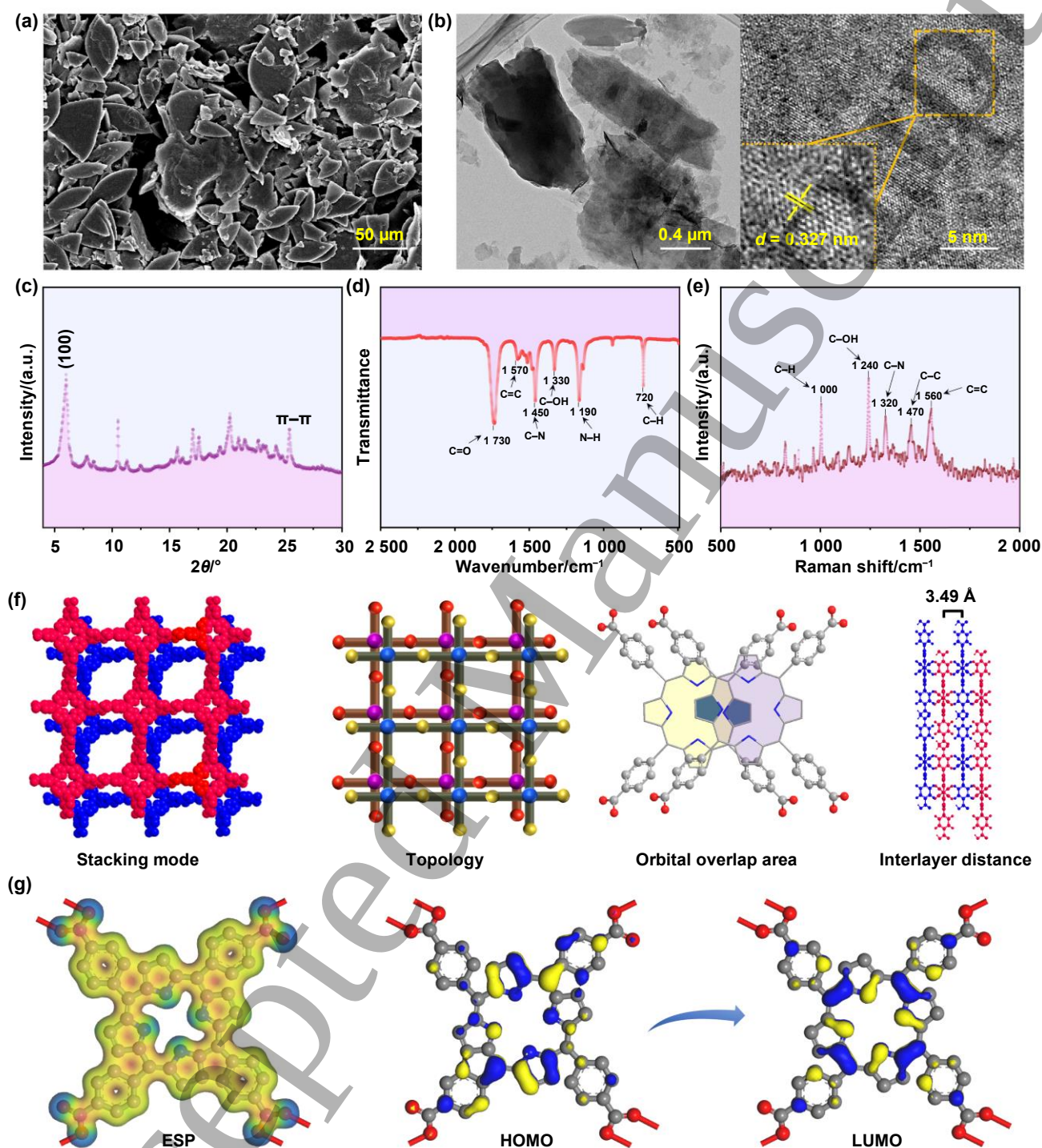


Figure 4. Characterization and structural display of TCPP-HOF. (a) SEM image of TCPP-HOF. (b) TEM and HRTEM images of TCPP-HOF. (c) XRD pattern of TCPP-HOF. (d) FT-IR spectrum of TCPP-HOF. (e) Raman spectrum of TCPP-HOF. (f) Schematics showing the stacking mode, topology, orbital overlap area, and interlayer distance of TCPP-HOF. (g) The calculated ESP, HOMO, and LUMO orbitals of TCPP monomer.

To decouple the contribution of the active layer from the electrode effects, we fabricated a control device with conventional electrodes to ascertain whether the synaptic performance originates intrinsically from the TCPP-HOF material rather than from interfacial hybridization with MXene. The control device features a platinum (Pt)/TCPP-HOF/indium tin oxide (ITO) structure, with an ITO substrate as the bottom electrode and thermally evaporated inert Pt as the top electrode. As shown in Figure 5(a), the I - V characteristics of this control device over 23 consecutive voltage sweeps ($0 \rightarrow 1.0 \rightarrow 0 \rightarrow -1.0 \rightarrow 0$ V) exhibit a conductance modulation trend similar to that of the pristine MXene/TCPP-HOF/MXene device (Figure 3(a)). The optimal PSC responses are achieved under identical pulse parameters: a positive pulse (8.0 V, 1 μ s, 2 μ s; Figure 5(b)) and a negative pulse (-11.0 V, 1 μ s, 9 μ s; Figure S8). In addition, the temporal evolution of the conductance in the control device closely matches that of the pristine device (Figures 5(c), S9, and S10). These collective results provide compelling evidence that the observed synaptic functionalities are intrinsic to the TCPP-HOF material, highlighting its role as a synaptic medium that operates independently of the electrode material.

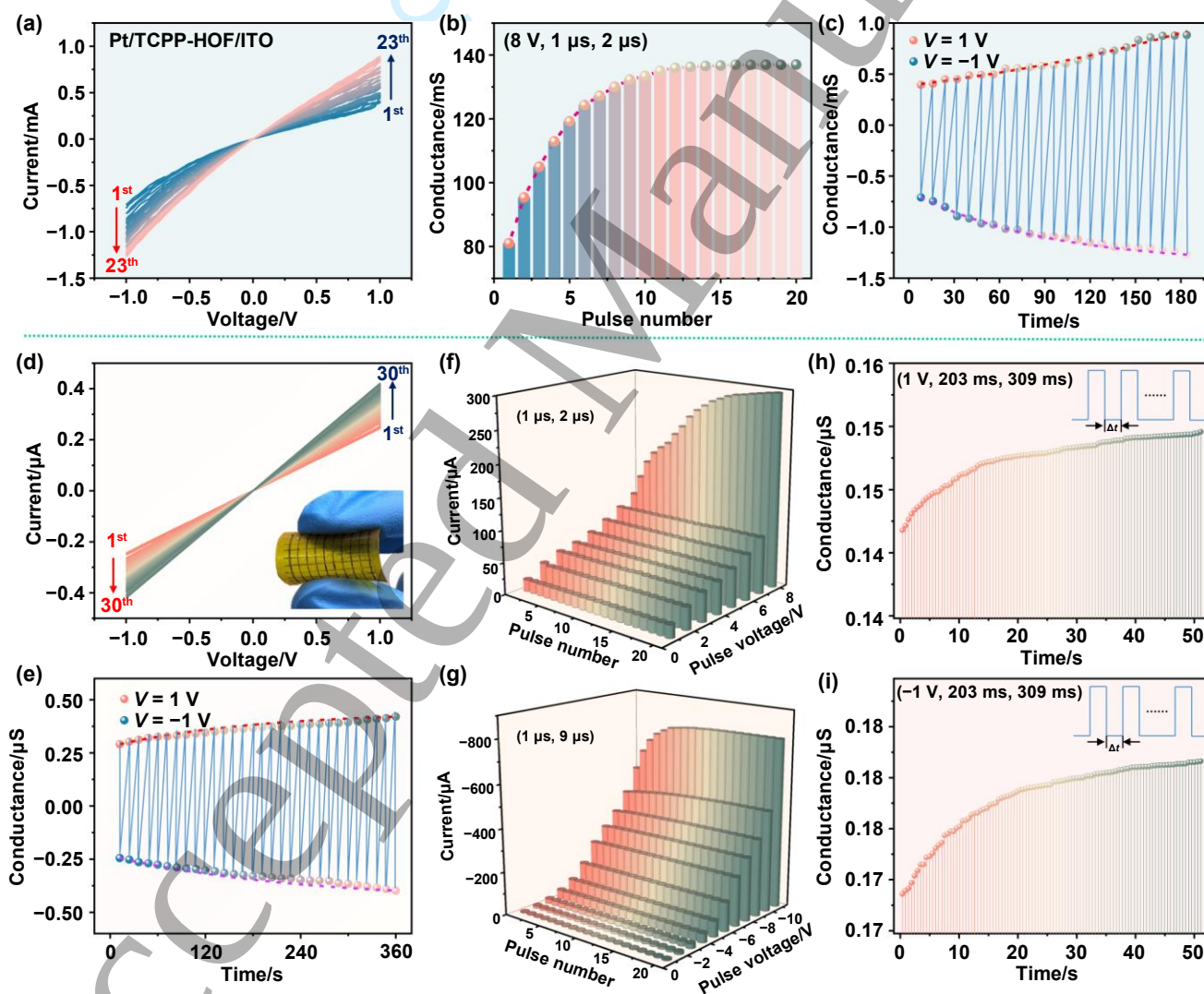


Figure 5. Electrical characterizations of the synaptic device under various electrode and bending states. (a) Cyclic I - V characteristics of the Pt/TCPP-HOF/ITO device under 23 consecutive voltage sweeps. (b) Conductance responses to pulse trains (amplitude: 8.0 V; width: 1 μ s; interval: 2 μ s). (c) Current variation

over time at ± 1.0 V. (d) Cyclic I - V characteristics of MXene/TCPP-HOF/MXene/PI after bending. (e) Current variation over time at ± 1.0 V after bending. (f) Positive PSC response with pulse amplitudes from 1.0 V to 9.0 V. (g) Negative PSC response with pulse amplitudes from -1.0 V to -11 V. (h) and (i) Conductance evolution under 1.0 V (h) and -1.0 V (i) pulse stimulations.

Moreover, we explored the mechanical flexibility of the MXene/TCPP-HOF/MXene/PI device. The all-solution-processability and PI substrate endow the device with excellent bendability. To assess the robustness of this flexibility, we subjected the device to 200 bending cycles (a bending radius of ~ 5 mm and a bending angle of $\sim 120^\circ$) and subsequently re-evaluated its electrical properties. The cyclic I - V characteristics ($0 \rightarrow 1.0 \rightarrow 0 \rightarrow -1.0 \rightarrow 0$ V) over 30 cycles remain largely unchanged after bending, with no significant performance degradation (Figure 5(d)). The device exhibits stable current levels with monotonic progression under the applied read voltages (Figure 5(e)) and maintains its responsive behavior under both positive (Figure 5(f)) and negative (Figure 5(g)) pulse trains. The time-dependent conductance under constant spikes (Figure 5(h) and (i)) is consistent with the pristine state. The device flexibility derives from the mechanical toughness of the self-supporting PI substrate, the hydrogen bonds of the TCPP-HOF active layers, and the solution-processable MXene electrodes. Unlike metal electrodes, this metal-electrode-free device maintains a stable in-device charge transport pathway under repeated mechanical deformation. Moreover, the ultrathin and lightweight PI-based array conformally adheres to human skin or textiles, holding the potential to implement energy-efficient and wearable intelligent sensing systems.

To evaluate its neuromorphic computing potential, we conducted CNN-based image recognition training using the TCPP-HOF device, leveraging its inherent synaptic capabilities [49-51]. As illustrated in Figure 6(a), the input images are first processed by a convolutional layer for feature extraction, followed by a pooling layer that reduces data dimensionality to mitigate overfitting. After these processing stages, the data are fed into a fully connected layer to generate the final output. The network, which comprises 1.2×10^5 synapses derived from 28×28 pixel MNIST images, features 10 output neurons corresponding to digits "0-9". Figure 6(b) depicts the crossbar device array implementing the fully connected layer, where the input and output neurons are represented by rows and columns, respectively. The synaptic weight values are represented by the device conductance, as defined in Equation (1).

$$I_x = \sum G_{xy} V_y \quad (1)$$

In the system, G_{xy} corresponds to the device conductance at (x, y) . The output value of neuron y is denoted as V_y , and the total input current I_x to neuron x is given by the multiplication of the conductance matrix and the voltage vector. Each synaptic weight W_i is determined by the difference in conductance G_i between two synaptic devices according to Equation (2), thereby enabling the representation of both positive and negative weights [52-54].

$$W_i \propto G_i = G_i^+ - G_i^- \quad (2)$$

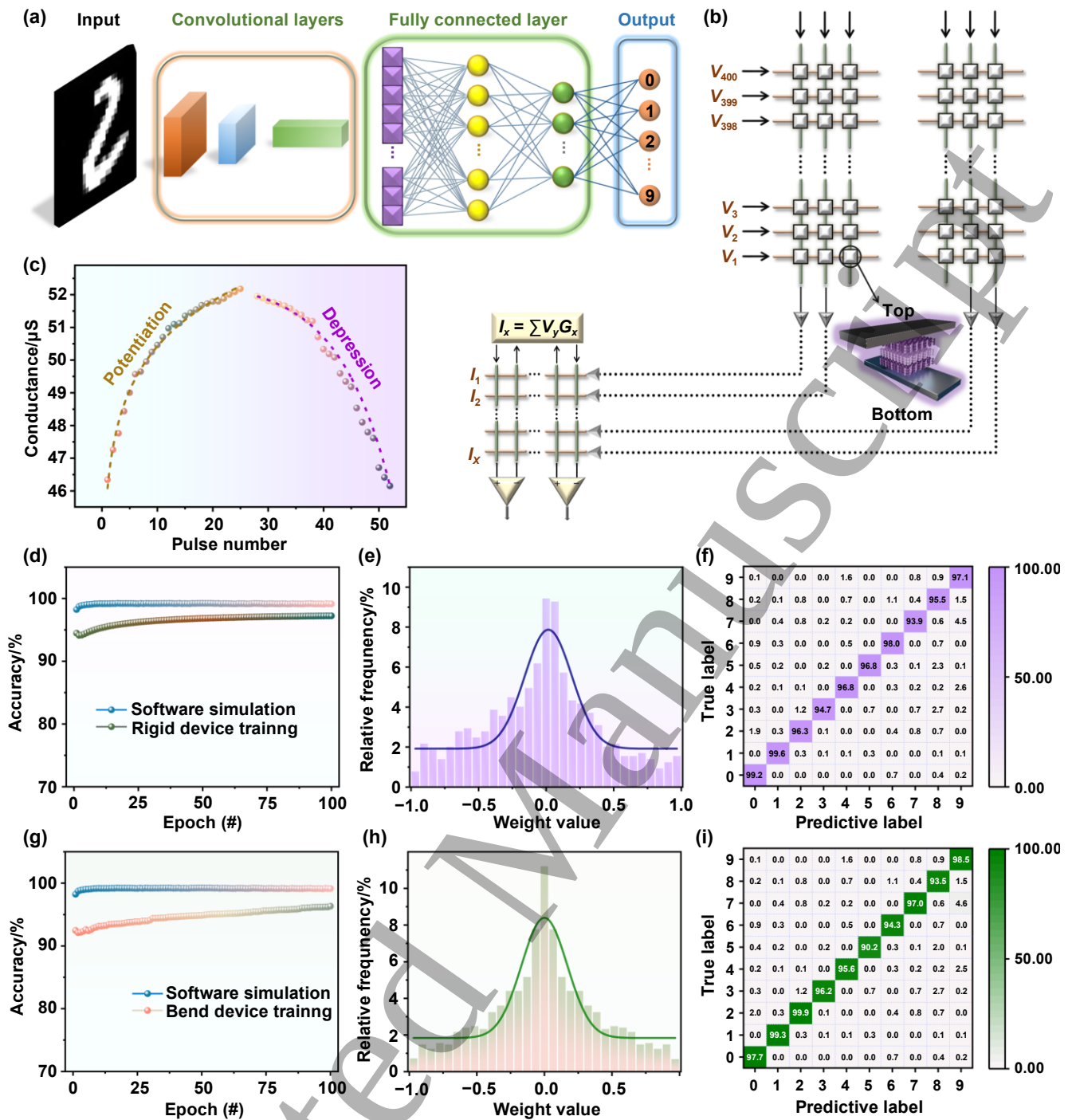


Figure 6. Neuromorphic computing based on CNN. (a) CNN architecture for MNIST recognition. (b) Synaptic device crossbar array. (c) Potentiation and depression of the rigid device. (d) and (g) Recognition accuracy achieved through software simulation and hardware training for the rigid device (d) and bent device (g). (e) and (h) Statistics of the weight distribution modulated by voltage pulses for the rigid device (e) and bent device (h). (f) and (i) Confusion matrices for evaluating the recognition accuracy for all digits “0–9” for the rigid device (f) and bent device (i).

The crossbar circuit executes the core synaptic operation through a grounded output line, enabling efficient weighted summation. The synaptic weights (W_i) are then updated via SET and RESET pulses within the learning algorithm [55-56]. As evidenced in Figure 6(c), the artificial synapse exhibits stable conductance

modulation, demonstrating clear potentiation and depression behaviors. The MNIST dataset-based recognition accuracy comparisons between the software simulation and our device are evaluated (Figure 6(d)). After iterative training, the network achieves a maximum accuracy of 97.23%, closely approaching the software baseline of 99.24%. Furthermore, Figure 6(e) illustrates the distribution of the voltage-pulse-modulated weights; its normal fitting confirms a broad conductance mapping range, manifesting the ability of the device to emulate software-derived weight matrices. The corresponding confusion matrix in Figure 6(f) reveals strong consistency between the predicted and actual labels for all digits “0–9”, with each category attaining recognition accuracies above 93%. To evaluate the effects of the cyclic bending operations on the CNN-based computing capacity, the MNIST recognition accuracy of the device is re-evaluated after 200 conformal deformation cycles, which also results in reliable potentiation and depression plasticity (Figure S11). The MNIST-based training results show no significant change before and after bending, with the maximum value remaining at 96.33% (Figure 6(g) and (h)). For the confusion matrix of digits “0–9” under bent condition, each category maintains a faithful recognition accuracy over 90% (Figure 6(i)). These findings highlight the potential of this all-solution-processable flexible device for enabling large-scale, high-density organic synaptic arrays and integrated neural-network circuits, paving a viable path toward bionic intelligent systems.

3. Conclusion

In summary, we developed a TCPP-HOF-pioneered all-solution-processable, degradable, and flexible synaptic emulator. The TCPP-HOF device achieves gradient conductance switching for bio-synaptic mimicking, including long-term potentiation (LTP), long-term depression (LTD), paired-pulse facilitation (PPF), and post-tetanic potentiation (PTP), implementing state-of-the-art HOF-based neuromorphic device with an ultrafast switching speed of 26 ns. The pulse stimulations achieve multilevel resistance states, which holds the potential for realizing high-density data storage. In addition, its environmentally friendly characteristics and degradability help to reduce the electronic burden and develop transient electronics. Moreover, the TCPP-HOF bio-inspired device achieves CNN-based computation with a digit recognition accuracy as high as 97.23%, which provides a new direction for the development of highly efficient organic intelligent sensing, cognitive computing, and neuromorphic systems.

Supporting information

Supporting information for this article is available online.

Acknowledgments

This work acknowledges financial support from the Natural Science Foundation of Jiangsu Province (BK20230074), the Young Elite Scientists Sponsorship Program by the China Association for Science and Technology (YESS20230439), the National Natural Science Foundation of China (62304148), the Key Technology Research Program of Suzhou (SYG2025132), and Research in Cutting-Edge Technology of

Suzhou (SYG202351). Q.Z. acknowledges the financial support from the City University of Hong Kong (7020148; 9239116; 9240189; 9380117; 9678403; 9680375; R-IND26401 and R-IND26402) and Hong Kong Branch of National Precious Metals Material Engineering Research Center (NPMR), Hong Kong, P. R. China.

Conflicts of interest

The authors declare that they have no conflicts of interest.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- [1] Qiu J W *et al.* 2024. Advancements in nanowire-based devices for neuromorphic computing: a review. *ACS Nano* **18**, 31632-31659.
- [2] Yao J, Teng Y, Wang Q N, He Y Q, Liu L W, Zhao C and Kang L X. 2025. Advancing intelligent neuromorphic computing: recent progress in all-optical-controlled artificial synaptic devices. *ACS Nano* **19**, 26320-26346.
- [3] Zhu Z Q, Shui J H, Wang T Y and Meng J L. 2026. Mechanical properties analysis of flexible memristors for neuromorphic computing. *Nano-Micro Lett.* **18**, 2.
- [4] Luo Z D, Zhang S Q, Liu Y, Zhang D W, Gan X T, Seidel J, Liu Y, Han G Q, Alexe M and Hao Y. 2022. Dual-ferroelectric-coupling-engineered two-dimensional transistors for multifunctional in-memory computing. *ACS Nano* **16**, 3362-3372.
- [5] Shao H *et al.* 2025. Adaptive in-sensor computing for enhanced feature perception and broadband image restoration. *Adv. Mater.* **37**, 2414261.
- [6] Wang Y X *et al.* 2026. Dual-mode optoelectronic neuromorphic memory for complex edge detection and recognition. *Adv. Funct. Mater.* **36**, e14949.
- [7] Pazos S, Zhu K C, Villena M A, Alharbi O, Zheng W W, Shen Y Q, Yuan Y, Ping Y and Lanza M. 2025. Synaptic and neural behaviours in a standard silicon transistor. *Nature* **640**, 69-76.
- [8] Zhang C, Chen M H, Pan Y L, Li Y, Wang K B, Yuan J W, Sun Y Q and Zhang Q C. 2023. Carbon nanodots memristor: an emerging candidate toward artificial biosynapse and human sensory perception system. *Adv. Sci.* **10**, 2207229.
- [9] Falcón-Moya R *et al.* 2020. Astrocyte-mediated switch in spike timing-dependent plasticity during hippocampal development. *Nat. Commun.* **11**, 4388.
- [10] Baek S, Jeong S, Ban H W, Ryu J, Kim Y, Gu D H, Son C, Yoon T S, Lee J and Son J S. 2023. Nanoscale vertical resolution in optical printing of inorganic nanoparticles. *ACS Nano* **17**, 24268-24281.
- [11] Bryan A C, Meyers J K, Serafin L Y, Krajnak Z A, Gentile R J and Cahoon J F. 2025. Metal organic

- chemical vapor deposition of hybrid perovskites. *J. Am. Chem. Soc.* **147**, 10212-10218.
- [12] Guo P J, Stoumpos C C, Mao L L, Sadasivam S, Ketterson J B, Darancet P, Kanatzidis M G and Schaller R D. 2018. Cross-plane coherent acoustic phonons in two-dimensional organic-inorganic hybrid perovskites. *Nat. Commun.* **9**, 2019.
- [13] Ni M J *et al.* 2025. Nanocolumn floating gates induced by the supramolecular blending strategy for highly improved charge trapping capacity. *ACS Appl. Mater. Interfaces* **17**, 44748-44756.
- [14] Li W H, Liu Q, Zhang Y N, Li C, He Z F, Choy W C H, Low P J, Sonar P and Kyaw A K K. 2020. Biodegradable materials and green processing for green electronics. *Adv. Mater.* **32**, 2001591.
- [15] Liu H C, Jian R R, Chen H B, Tian X L, Sun C L, Zhu J, Yang Z G, Sun J Y and Wang C S. 2019. Application of biodegradable and biocompatible nanocomposites in electronics: current status and future directions. *Nanomaterials (Basel)* **9**, 950.
- [16] Ou Q Q, Yang B, Zhang J Y, Liu D P, Chen T Q, Wang X, Hao D D, Lu Y and Huang J. 2021. Degradable photonic synaptic transistors based on natural biomaterials and carbon nanotubes. *Small* **17**, 2007241.
- [17] Li B, Gao D P, Zhang C L, Yu Z X, Stolterfoht M, Lin Y H, Lenz M, Snaith H J and Zhu Z L. 2026. Closed-loop manufacturing for sustainable perovskite photovoltaics. *Nat. Rev. Mater.* **11**, 10-25.
- [18] Ding M Y, Jiang T, Wang B, Li Y S, Zhang J Y, Huang J, Ji D Y and Hu W P. 2024. Environmentally friendly and degradable organic neuromorphic vision sensors. *Matter* **7**, 1736-1749.
- [19] He X T, Luo Y H, Zheng Z Y, Wang C, Wang J Y, Hong D L, Zhai L H, Guo L H and Sun B W. 2019. Porphyrin-based hydrogen-bonded organic frameworks for the photocatalytic degradation of 9,10-diphenylanthracene. *ACS Appl. Nano Mater.* **2**, 7719-7727.
- [20] Zhao X, Yin Q, Mao X N, Cheng C, Zhang L, Wang L, Liu T F, Li Y Y and Li Y G. 2022. Theory-guided design of hydrogen-bonded cobaltoporphyrin frameworks for highly selective electrochemical H₂O₂ production in acid. *Nat. Commun.* **13**, 2721.
- [21] Yin Q *et al.* 2022. Metallization-prompted robust porphyrin-based hydrogen-bonded organic frameworks for photocatalytic CO₂ reduction. *Angew. Chem. Int. Ed.* **61**, e202115854.
- [22] Xiong H L, Ling S T, Li Y, Duan F, Zhu H, Lu S L and Du M L. 2022. Flexible and recyclable bio-based transient resistive memory enabled by self-healing polyimine membrane. *J. Colloid Interface Sci.* **608**, 1126-1134.
- [23] Wang G L, Sun F, Zhou S Y, Zhang Y Z, Zhang F, Wang H Y, Huang J J and Zheng Y. 2024. Enhanced memristive performance via a vertically heterointerface in nanocomposite thin films for artificial synapses. *ACS Appl. Mater. Interfaces* **16**, 12073-12084.
- [24] Jaafar A H, Al Habsi S K S, Braben T, Venables C, Francesconi M G, Stasiuk G J and Kemp N T. 2024. Unique coexistence of two resistive switching modes in a memristor device enables multifunctional neuromorphic computing properties. *ACS Appl. Mater. Interfaces* **16**, 43816-43826.
- [25] Chen J L, Chiang T C and Liu P T. 2025. All-metal-oxide heterojunction optoelectronic synapses with multilevel memory for artificial visual perception applications. *Small* **21**, 2502271.

- [26] Zhang C, Li Y, Yu F, Wang G, Wang K B, Ma C L, Yang X B, Zhou Y and Zhang Q C. 2023. Visual growth of nano-HOFs for low-power memristive spiking neuromorphic system. *Nano Energy* **109**, 108274.
- [27] Wang S Y *et al.* 2019. A MoS₂/PTCDA hybrid heterojunction synapse with efficient photoelectric dual modulation and versatility. *Adv. Mater.* **31**, 1806227.
- [28] Dong L Y, Xue B J, Wei G D, Yuan S, Chen M, Liu Y, Su Y, Niu Y, Xu B S and Wang P. 2024. Highly promising 2D/1D BP-C/CNT bionic opto-olfactory co-sensory artificial synapses for multisensory integration. *Adv. Sci.* **11**, 2403665.
- [29] Sung J, Kim S W, Lee D, Moon S, Lee E and Kim H H. 2025. Co-stimuli-driven 2D WSe₂ optoelectronic synapses for neuromorphic computing. *Small* **21**, 2504024.
- [30] Xu Z *et al.* 2025. Preparation of organic N-fused perylenediimide-MXene hybrid material for robust versatile memristive device. *Int. J. Extrem. Manuf.* **7**, 025507.
- [31] Li Y, Ling S T, He R Y, Zhang C, Dong Y, Ma C L, Jiang Y C, Gao J, He J H and Zhang Q C. 2023. A robust graphene oxide memristor enabled by organic pyridinium intercalation for artificial biosynapse application. *Nano Res.* **16**, 11278-11287.
- [32] Li J Y *et al.* 2023. Polymeric memristor based artificial synapses with ultra-wide operating temperature. *Adv. Mater.* **35**, 2209728.
- [33] Li Y X *et al.* 2025. A synaptic device based on a coordinated-metal-synergetic covalent organic framework for highly efficient neuromorphic application. *Chem. Mater.* **37**, 5355-5364.
- [34] Zhang C *et al.* 2022. Nanofiber architecture engineering implemented by electrophoretic-induced self-assembly deposition technology for flash-type memristors. *ACS Appl. Mater. Interfaces* **14**, 3111-3120.
- [35] Zhang W J, Li Y X, Zhang C, Yuan J W, Wang Y Y, Cheng X L, Qian W H, Sun S X and Li Y. 2024. Synchronous regulation of hydrophobic molecular architecture and interface engineering for robust WORM-type memristor. *Adv. Funct. Mater.* **34**, 2404625.
- [36] Wang T Y, Meng J L, Li Q X, He Z Y, Zhu H, Ji L, Sun Q Q, Chen L and Zhang D W. 2021. Reconfigurable optoelectronic memristor for in-sensor computing applications. *Nano Energy* **89**, 106291.
- [37] Choi S, Yang J and Wang G. 2020. Emerging memristive artificial synapses and neurons for energy-efficient neuromorphic computing. *Adv. Mater.* **32**, 2004659.
- [38] Wu Y Y, Xu G Y, Yan Z H, Zhou X Y, Yang F L, Dai J W, Wu S Q, Li Z Y and Sato O. 2025. Altered mode of structural changes in solid solutions leading to dual modulation: spin transition temperatures and steps. *J. Am. Chem. Soc.* **147**, 14401-14410.
- [39] Zhao Y Y, Sun W J, Wang J, He J H, Li H, Xu Q F, Li N J, Chen D Y and Lu J M. 2020. All-Inorganic ionic polymer-based memristor for high-performance and flexible artificial synapse. *Adv. Funct. Mater.* **30**, 2004245.
- [40] Zhang Z S and Yan B. 2023. Convolution neural network-assisted smart fluorescent-tongue based on lanthanide ion-induced forming MOF/HOF composite for differentiation of flavor compounds and

- wine identification. *ACS Sens.* **8**, 3585-3594.
- [41] Yang X, Huang J, Gao S Y, Zhao Y Q, Huang T, Li H F, Liu T F, Yu Z Y and Cao R. 2023. Solution-processed hydrogen-bonded organic framework nanofilms for high-performance resistive memory devices. *Adv. Mater.* **35**, 2305344.
- [42] Song K Y, Yang H L, Chen B J, Lin X L, Liu Y, Liu Y Z, Li H H, Zheng S T and Chen Z R. 2023. The facile implementing ternary resistive memory in graphite-like melamine-cyanuric acid hydrogen-bonded organic framework with high ternary yield and environmental tolerance. *Appl. Surf. Sci.* **608**, 155161.
- [43] Chen S M *et al.* 2024. Multistate structures in a hydrogen-bonded polycatenation non-covalent organic framework with diverse resistive switching behaviors. *Nat. Commun.* **15**, 298.
- [44] Hu S W, Zhao H, Liang M, Zhou N N, Ding B, Liu X L, Zeng Y F, Tang B, Hao J J and Xue P C. 2024. Luminescent porous organic crystals for adsorptive separation of toluene and methylcyclohexane. *ACS Appl. Mater. Interfaces* **16**, 4863-4872.
- [45] Chen G S, Huang S M, Ma X M, He R W and Ouyang G F. 2023. Encapsulating and stabilizing enzymes using hydrogen-bonded organic frameworks. *Nat. Protoc.* **18**, 2032-2050.
- [46] Tian H *et al.* 2017. Emulating bilingual synaptic response using a junction-based artificial synaptic device. *ACS Nano* **11**, 7156-7163.
- [47] Burlingham S R *et al.* 2022. Induction of synapse formation by *de novo* neurotransmitter synthesis. *Nat. Commun.* **13**, 3060.
- [48] Lee Y *et al.* 2024. Programmable retention characteristics in MoS₂-based atomistors for neuromorphic and reservoir computing systems. *ACS Nano* **18**, 14327-14338.
- [49] Yao P, Wu H Q, Gao B, Tang J S, Zhang Q T, Zhang W Q, Yang J J and Qian H. 2020. Fully hardware-implemented memristor convolutional neural network. *Nature* **577**, 641-646.
- [50] LeCun Y, Bengio Y and Hinton G. 2015. Deep learning. *Nature* **521**, 436-444.
- [51] Esteva A, Robicquet A, Ramsundar B, Kuleshov V, Depristo M, Chou K, Cui C, Corrado G, Thrun S and Dean J. 2019. A guide to deep learning in healthcare. *Nat. Med.* **25**, 24-29.
- [52] Weilenmann C, Ziogas A N, Zellweger T, Portner K, Mladenović M, Kaniselman M, Moraitis T, Luisier M and Emboras A. 2024. Single neuromorphic memristor closely emulates multiple synaptic mechanisms for energy efficient neural networks. *Nat. Commun.* **15**, 6898.
- [53] Rao M Y *et al.* 2023. Thousands of conductance levels in memristors integrated on CMOS. *Nature* **615**, 823-829.
- [54] Han G, Lee C, Lee J E, Seo J, Kim M, Song Y B, Seo Y H and Lee D. 2021. Alternative negative weight for simpler hardware implementation of synapse device based neuromorphic system. *Sci. Rep.* **11**, 23198.
- [55] Feng X W, Li S F, Wong S L, Tong S, Chen L, Zhang P P, Wang L F, Fong X, Chi D Z and Ang K W. 2021. Self-selective multi-terminal memtransistor crossbar array for in-memory computing. *ACS Nano*

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

15, 1764-1774.

[56] Chen P H, Lin C Y, Chang T C, Eshraghian J K, Chao Y T, Lu W D and Sze S M. 2022. Investigating selectorless property within niobium devices for storage applications. *ACS Appl. Mater. Interfaces* **14**, 2343-2350.